

Anomalous electromagnetic moments of the tau lepton at the LHeC and LH μ C

Photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$ production

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Why anomalous electromagnetic moments of the τ lepton?



- For any charged lepton, the gyromagnetic factor is written as

$$g_\ell = 2(1 + a_\ell), \quad a_\ell = \frac{g_\ell - 2}{2}.$$

- The electron anomalous magnetic moment is measured with extraordinary precision, while the muon sector remains a long-standing arena for precision tests of quantum corrections and possible new physics.
- The τ lepton is much heavier and experimentally less constrained; if beyond-the-Standard-Model effects scale with lepton mass, the τ sector can be especially sensitive.
- Photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$ production provides a direct probe of the $\gamma\tau\tau$ vertex and therefore of a_τ and d_τ .

Physics target of this talk

Compare the truth-level sensitivity of the **LHeC** and the **LH μ C** to anomalous τ electromagnetic moments using a benchmark $\delta a_\tau = 0.008$ in photon-induced τ -pair production.

Current experimental status of a_τ



- Earlier direct constraints came from OPAL, L3, and DELPHI in the LEP era.
- Heavy-ion measurements by ATLAS Pb+Pb and CMS Pb+Pb extended the reach using photon-induced processes at low di- τ masses.
- The recent CMS pp analysis observed $\gamma\gamma \rightarrow \tau^+\tau^-$ and set significantly improved limits on a_τ and d_τ [Rept. Prog. Phys. 87 (2024) 107801].
- The present future-collider study is motivated by the same central idea: **the BSM effect becomes more visible in the hard region of the mass spectrum.**

Representative CMS pp context

Observed significance $\sim 5.3\sigma$; 95% CL interval roughly $-0.0042 < a_\tau < 0.0062$; complementary constraints also obtained on d_τ .

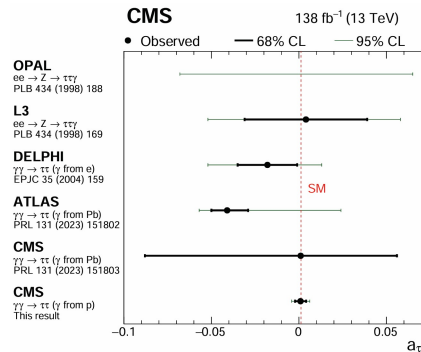


Figure: World-comparison plot for the anomalous tau magnetic moment a_τ , including LEP, heavy-ion, and CMS pp constraints [Rept. Prog. Phys. 87 (2024) 107801].

Why photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$?



- The $\gamma\gamma \rightarrow \tau^+\tau^-$ process probes the $\gamma\tau\tau$ interaction directly and cleanly.
- In hadron or lepton-hadron environments, equivalent-photon emission from the beams can produce a relatively simple electromagnetic initial state.
- The invariant-mass spectrum of the $\tau^+\tau^-$ pair is particularly powerful because **anomalous dipole-type couplings harden the high-mass tail**.
- This makes the process attractive both for present measurements and for future-machine sensitivity studies (FCC, LHeC, LH μ C, ...).

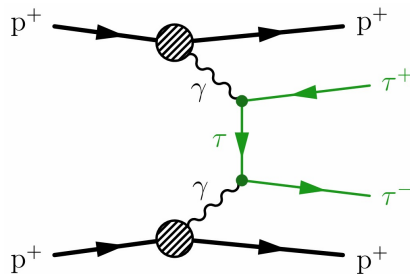


Figure: Representative topology for photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$ production in proton-based collisions.

Take-home message

The most important discriminating observable is the hard tail of the $\tau^+\tau^-$ invariant-mass distribution.

General electromagnetic vertex and dipole moments



- The most general electromagnetic vertex of a charged lepton can be parameterized in terms of form factors:

$$\Gamma^\mu(q^2) = \gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m_\tau} F_2(q^2) + \sigma^{\mu\nu} q_\nu \gamma_5 F_3(q^2).$$

- At zero momentum transfer, the static electromagnetic moments are identified through

$$F_2(0) = a_\tau, \quad F_3(0) = -\frac{2m_\tau}{e} d_\tau.$$

Interpretation

a_τ measures the anomalous magnetic dipole moment, while d_τ measures the electric dipole moment and is CP-odd.

Low-energy dipole interaction for the $\gamma\tau\tau$ vertex



- At the effective-interaction level, the anomalous electromagnetic moments are captured by a dipole-type term in the Lagrangian:

$$\mathcal{L}_{\tau\tau\gamma} = \frac{1}{2} \bar{\tau} \sigma^{\mu\nu} \left(a_{\tau} \frac{e}{2m_{\tau}} - i d_{\tau} \gamma_5 \right) \tau F_{\mu\nu}.$$

- The magnetic contribution is CP-even and the electric-dipole contribution is CP-odd.
- In a collider process such as $\gamma\gamma \rightarrow \tau^+\tau^-$, these couplings change both the overall rate and the shape of kinematic distributions.
- In practice, the mass spectrum is the cleanest place to see the corresponding BSM distortion.

Take-home message

The talk focuses on the benchmark $\delta a_{\tau} = 0.008$, keeping the emphasis on the magnetic-moment sensitivity.

SMEFT interpretation of anomalous τ moments



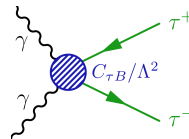
- In the SMEFT language, the relevant dim-6 operators can be written as

$$\mathcal{L}_{\text{BSM}} = \bar{L}_\tau \sigma^{\mu\nu} \tau_R H \left[\frac{C_{\tau B}}{\Lambda^2} B_{\mu\nu} + \frac{C_{\tau W}}{\Lambda^2} W_{\mu\nu} \right] + \text{h.c.}$$

- After electroweak symmetry breaking, the physical photon combination appears through a linear combination of the Wilson coefficients.
- The real part gives the CP-even magnetic deformation and the imaginary part gives the CP-odd electric-dipole deformation.

$$\delta d_\tau = \frac{\sqrt{2} v}{\Lambda^2} \text{Im}[\cos \theta_W C_{\tau B} - \sin \theta_W C_{\tau W}],$$

$$\delta a_\tau = \frac{2m_\tau}{e} \frac{\sqrt{2} v}{\Lambda^2} \text{Re}[\cos \theta_W C_{\tau B} - \sin \theta_W C_{\tau W}].$$



Dipole-type

SMEFT insertion
associated with
 $C_{\tau B}/\Lambda^2$.

Benchmark implementation used in this study



- Event generation is performed with the `SMEFTsim_top_alphaScheme_UFO` model.
- The present local benchmark corresponds to a nonzero CP-even dipole deformation associated with the tau sector and interpreted as $\delta a_\tau = 0.008$.
- In the local banners, the BSM benchmark is realized using a nonzero tau dipole Wilson-coefficient entry and $\Lambda = 2$ TeV for the EFT normalization.
- The study is designed as a **truth-level comparison** between the SM and a single illustrative BSM benchmark, not yet as a detector-level limit extraction.

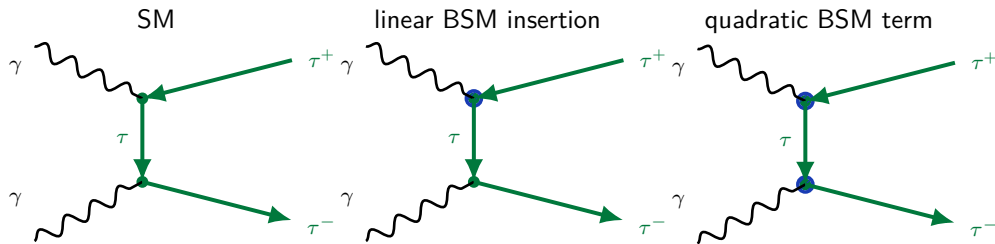
Color legend

- $a_\tau, \delta a_\tau$: magnetic moment sector
- $d_\tau, \delta d_\tau$: electric dipole sector
- C_{TB}, C_{TW}, Λ : EFT parameters

SMEFTsim

- GitHub: <https://github.com/SMEFTsim/SMEFTsim>
- Reference: <https://arxiv.org/abs/2012.11343>

Amplitude structure and representative diagrams



Interpretation of the generated samples

The SM sample is generated with $NP=0$. The benchmark BSM sample is generated with $NP \leq 2$, so the prediction includes the Standard Model contribution together with the interference term and the quadratic EFT contribution in the MadGraph operator-order convention used in this study.

Future lepton-hadron collider program considered here



LHeC

- proton beam: $E_p = 7$ TeV
- electron beam: $E_e = 50$ GeV
- center-of-mass energy: $\sqrt{s} \approx 1.2$ TeV

LH μ C

- proton beam: $E_p = 7$ TeV
- antimuon / μ^+ beam in the local MadGraph setup: $E_\mu = 500$ GeV
- center-of-mass energy: $\sqrt{s} \approx 3.7$ TeV

- The present study keeps the comparison symmetric: the same process, the same EFT benchmark, and the same LHE-level analysis logic are used for both colliders.
- This makes the difference in the observed BSM distortion directly interpretable as a collider-environment effect within the current setup.

MadGraph process definition and sample types



SM generation

```
import model SMEFTsim_top_alphaScheme_UFO
generate a a > ta+ ta- / h h1 NP=0
SMHLOOP=0
```

BSM benchmark generation

```
import model SMEFTsim_top_alphaScheme_UFO
generate a a > ta+ ta- / h h1 NP<=2
SMHLOOP=0
```

- The hard process is identical in the two colliders; only the beam configuration changes.
- NP=0 isolates the SM contribution.
- NP<=2 includes the benchmark BSM effects together with the SM piece in the operator-order counting used in the model.
- This is especially important in the hard tail, where quadratic contributions can visibly enhance the spectrum.

Beam configuration and run-card logic



LHeC banner essentials

```
lpp1 = 2  
lpp2 = 3  
ebeam1 = 7000.0  
ebeam2 = 50.0  
pdlabel1 = iww  
pdlabel2 = iww
```

LHmuC banner essentials

```
lpp1 = 2  
lpp2 = -4  
ebeam1 = 7000.0  
ebeam2 = 500.0  
pdlabel1 = iww  
pdlabel2 = iww
```

- The current local setup uses an equivalent-photon description of the beams through the `iww` choice.
- This is therefore a generator-level truth study using an effective photon-emission picture, not a full detector simulation.

Generation cuts, benchmark choice, and statistics



Run-card highlights

```
mmll = 10.0 GeV
ptl = -1.0
etal = -1.0
drll = -1.0
```

- The generation is largely inclusive, with a minimal invariant-mass cut $m_{\ell\ell} > 10$ GeV.
- No strong truth-level acceptance cuts are imposed on lepton transverse momentum, rapidity, or separation at generation.

Sample statistics

- 5 independent runs per sample
- 1M unweighted events per run
- 5M events per collider and per hypothesis
- LHE header cross sections combined with event-count weighting

Benchmark

Present comparison: $\delta a_\tau = 0.008 @ \sqrt{s} = 2$ TeV at truth level.

Benchmark cross sections and Monte Carlo luminosities



Sample	σ_{SM} [pb]	σ_{BSM} [pb]	N_{events}
LHeC@1.2 TeV	50.695	51.923	5×10^6
LH μ C@3.7 TeV	55.094	56.461	5×10^6

- The effective luminosities inferred from the Monte Carlo sample sizes are approximately 97.5 fb^{-1} for the LHeC benchmark and 89.7 fb^{-1} for the LH μ C benchmark.
- These values are **Monte Carlo effective luminosities** used for the plot normalization logic; they are not the final projected machine luminosities for a full experimental analysis.

Truth-level analysis workflow implemented in the local code



- ① Read all `run_*` LHE or LHE.GZ files under the corresponding Events/ directory.
- ② Parse final-state τ^+ and τ^- four-momenta from each event.
- ③ Construct the pair observables $M_{\tau^+\tau^-}$ and $Y_{\tau^+\tau^-}$.
- ④ Convert binned event counts into weighted differential spectra using the LHE cross section.
- ⑤ Build the physical BSM/SM ratio from $(d\sigma/dM)_{\text{BSM}}$ and $(d\sigma/dM)_{\text{SM}}$.
- ⑥ Also build legacy count-ratio plots to keep continuity with earlier visualizations.
- ⑦ Fit the ratio using a linear ansatz $y(M) = aM + b$ to summarize tail hardening.

Interpretation of the fit slope

A positive slope indicates that the BSM enhancement grows toward larger $M_{\tau^+\tau^-}$, i.e. the anomalous coupling hardens the invariant-mass tail.

Observables used in the comparison



Primary observables

$$\frac{d\sigma}{dM_{\tau^+\tau^-}} \left(\frac{pb}{GeV} \right), \quad \frac{d\sigma}{dY_{\tau^+\tau^-}} \left(\frac{pb}{GeV} \right),$$

- The invariant-mass spectrum is expected to be the most sensitive observable.
- The rapidity distribution is useful as a complementary shape diagnostic.

Ratio observables

$$\frac{(d\sigma/dM)_{BSM}}{(d\sigma/dM)_{SM}}$$

- Physical weighted ratio: directly compares the differential spectra.
- Legacy ratio plots: retain continuity with the earlier plotting style and support compact linear-fit summaries.

Take-home message

The entire comparison is designed to quantify how strongly the benchmark $\delta a_\tau = 0.008$ distorts the high-mass shape relative to the Standard Model.

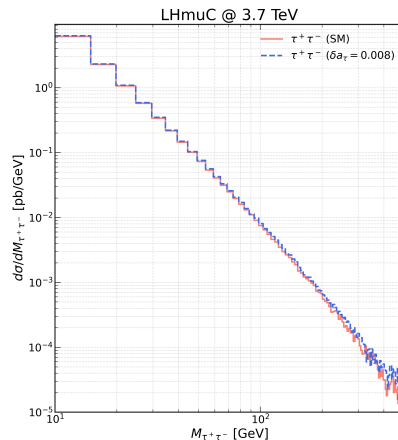
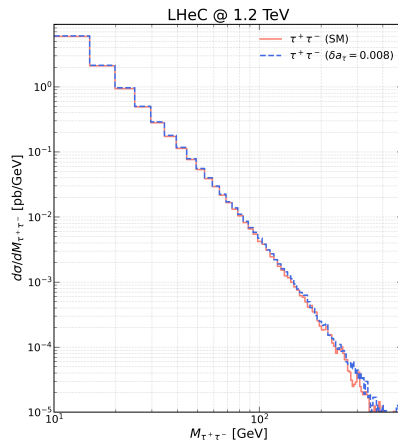
Invariant-mass spectrum: LHeC vs LH μ C

Figure: Comparison of the invariant-mass distributions for photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$ production at the LHeC and LH μ C for the SM and the benchmark anomalous coupling $\delta a_\tau = 0.008$. In both cases, the BSM contribution hardens the $M_{\tau^+\tau^-}$ spectrum relative to the SM expectation.

Findings from the invariant-mass comparison

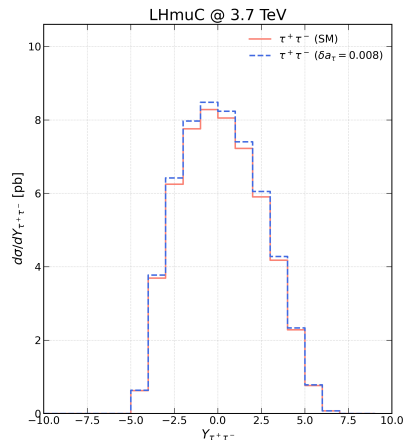
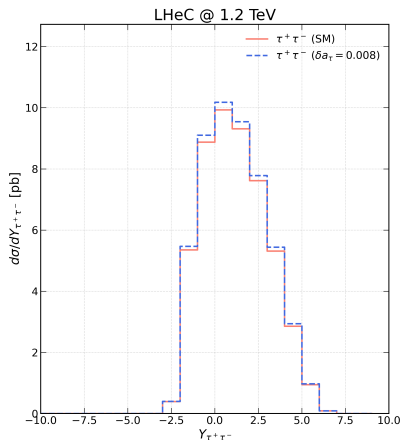


- In both collider configurations, the benchmark value $\delta a_\tau = 0.008$ produces a harder $M_{\tau^+\tau^-}$ spectrum than the Standard Model prediction.
- The separation between the SM and BSM curves becomes increasingly visible toward larger invariant mass, showing that the anomalous dipole contribution mainly affects the hard tail.
- This trend is physically expected, since dipole-type interactions become more important at higher momentum transfer.
- The effect is more pronounced for the LH μ C, where the BSM enhancement in the high- $M_{\tau^+\tau^-}$ region is visibly stronger than in the LHeC case.
- Therefore, already at the level of the invariant-mass spectrum, the LH μ C appears more sensitive to anomalous τ -dipole effects in the present truth-level benchmark setup.

Take-home message

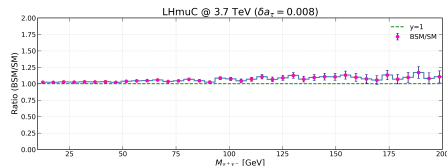
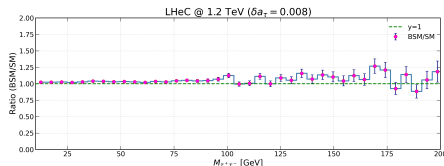
The invariant-mass tail is the most discriminating observable in this study, and its stronger deformation at the LH μ C already suggests a better sensitivity to a_τ than at the LHeC.

Rapidity distribution: LHeC vs LH μ C



- The LH μ C distribution is more central because its $p\mu^+$ beam-energy configuration is less asymmetric than the pe setup of the LHeC.

Weighted physical ratio: LHeC vs LH μ C

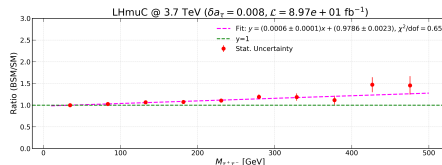
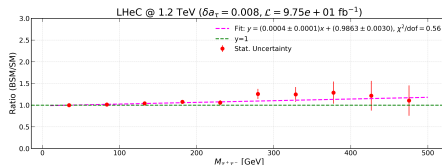


- This is the most directly physical comparison, because it uses the weighted differential spectra rather than only raw count ratios.
- For both colliders, the ratio stays close to unity at low mass and rises above unity as the invariant mass increases.
- The rise is stronger for the LH μ C, showing that the same benchmark δa_τ produces a larger relative deviation from the SM in that environment.

Take-home message

The physical ratio confirms the stronger high-mass BSM/SM deformation at the LH μ C.

Uniform-bin linear-fit comparison

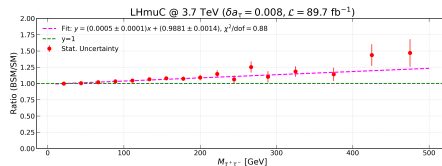
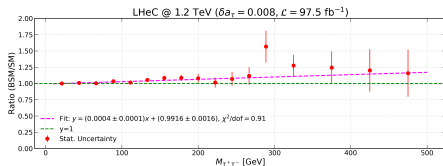


- The linear fit $y(M) = aM + b$ compresses the visual trend into a compact measure of tail hardening.
- In the present benchmark study, the fitted slope is larger for the LH μ C than for the LHeC.
- A larger positive slope means a more rapidly growing departure from the SM toward high invariant mass.

Take-home message

Even at the level of a compact slope summary, the LH μ C shows the stronger a_τ -sensitive growth with $M_{\tau^+\tau^-}$.

Custom-bin linear-fit comparison



- The custom-bin fit places more flexibility in the higher-mass region where the anomalous effect is most relevant.
- The qualitative conclusion remains stable: the LH μ C exhibits the steeper rise and therefore the stronger BSM sensitivity.
- The hierarchy between colliders is therefore robust across the direct spectrum plots, weighted ratios, and the two fit strategies.

Take-home message

Across all comparison views in this benchmark study, the LH μ C appears more sensitive to a_τ than the LHeC.

Combined interpretation of the LHeC vs LH μ C comparison

What all five comparisons consistently show

- The anomalous benchmark hardens the $\tau^+\tau^-$ invariant-mass spectrum.
- The rapidity spectrum changes much less strongly.
- The BSM/SM ratio grows with $M_{\tau^+\tau^-}$ rather than remaining flat.
- The growth is stronger for the LH μ C than for the LHeC.

Scientifically careful conclusion

Within the present **truth-level benchmark study** based on merged MadGraph LHE samples, the LH μ C appears more promising than the LHeC for probing a_τ through photon-induced $\tau^+\tau^-$ production.

- This statement concerns the current generator-level setup and should not yet be read as a final detector-level exclusion result.

How the future-collider study connects to CMS pp results



- CMS has observed $\gamma\gamma \rightarrow \tau^+\tau^-$ in pp collisions and used the rate and mass-shape information to constrain a_τ and d_τ .
- The analysis relies on an exclusivity strategy: low hadronic activity, low acoplanarity, and low track multiplicity near the dilepton or ditau vertex.
- The key conceptual overlap with our future-collider study is the same: BSM effects are enhanced in the higher-mass region.
- Future lepton-hadron colliders offer a complementary environment in which photon-induced processes can be studied with different beam kinematics and cleaner initial-state control.

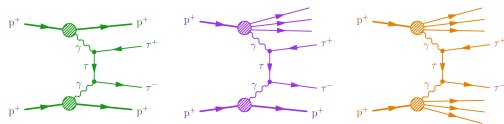


Figure: Schematic illustration of photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$ production in pp collisions, showing the elastic, semiexclusive, and dissociative topologies relevant for the CMS analysis.

Main conclusions



- We studied photon-induced $\gamma\gamma \rightarrow \tau^+\tau^-$ production at two future lepton-hadron colliders: the LHeC and the LH μ C.
- The benchmark anomalous coupling $\delta a_\tau = 0.008 @ \sqrt{s} = 2$ TeV hardens the invariant-mass spectrum relative to the Standard Model.
- The rapidity distributions do show a clear collider-dependent effect: the LH μ C distribution is more central than the LHeC one.
- The BSM/SM ratio grows with $M_{\tau^+\tau^-}$, showing that the hard tail is the key region for sensitivity.
- Across the invariant-mass, weighted-ratio, and fit-based comparisons, the LH μ C exhibits the stronger high-mass response.

Bottom line

In the present truth-level benchmark study, the LH μ C is favored over the LHeC for probing the anomalous magnetic moment a_τ in photon-induced τ -pair production.

Outlook



- Extend the study from truth level to **realistic detector-level** analyses.
- Implement acceptance and reconstruction effects for the relevant visible τ final states.
- Translate the present shape distortions into explicit projected constraints on a_τ at the **LHeC** and **LH μ C**.
- Generalize the benchmark study to the CP-odd sector and to simultaneous fits in the (C_{TB}, C_{TW}) plane.
- Explore luminosity scenarios, optimization of mass-window selections, and systematic uncertainties.
- Connect the future-collider projections more directly to the CMS pp strategy and other current measurements.



Backup: MG settings summary



Hard-process setup

```
import model SMEFTsim_top_alphaScheme_UF0
generate a a > ta+ ta- / h h1 NP=0 SMHLOOP=0
import model SMEFTsim_top_alphaScheme_UF0
generate a a > ta+ ta- / h h1 NP<=2 SMHLOOP=0
```

- Beam configurations: 7 TeV $\oplus$ 50 GeV for LHeC and 7 TeV $\oplus$ 500 GeV for LH μ C.
- Equivalent-photon setup through iww; minimal generation cut $m_{\ell\ell} > 10$ GeV.

Backup: EFT parameter summary



Quantity	Role in the study
$a_\tau, \delta a_\tau$	CP-even anomalous magnetic-moment sector
$d_\tau, \delta d_\tau$	CP-odd electric-dipole sector
C_{TB}, C_{TW}	SMEFT Wilson coefficients entering the tau dipole interaction
Λ	EFT scale used for normalization of the dim-6 operators
$M_{\tau^+\tau^-}$	primary sensitivity observable in the present benchmark study

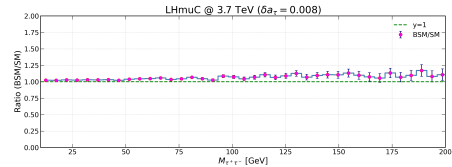
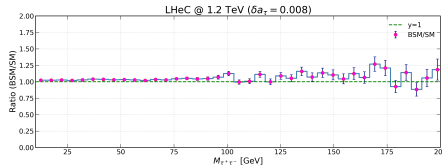
Backup: sample statistics summary



Collider	N_{runs}	$N_{\text{evts/run}}$	N_{tot}	$\mathcal{L}_{\text{eff}} [\text{fb}^{-1}]$
LHeC SM / BSM	5	10^6	5×10^6	~ 97.5
LH μ C SM / BSM	5	10^6	5×10^6	~ 89.7

- Cross sections are read from the LHE headers and combined with event-count weighting.
- The effective luminosities summarize the sample sizes used by the plotting and fitting workflow.

Backup: side-by-side weighted-ratio comparison



Take-home message

The weighted BSM/SM ratio remains the clearest visual summary of why the LH μ C appears more sensitive to a_τ in this benchmark study.